

11_popquiz Electric field ↕

⚠ This is a preview of the published version of the quiz

Started: Mar 22 at 7:10pm

Quiz Instructions

□



Question 1 25 pts

the electric field around an isolated positive charge is oriented

☐

inward

☐

the electric field has no direction

☐

It depends on the test charge

☐

outward



Question 2 25 pts

The direction of an electric field is @ right. An electron is placed in this electric field. The force acting on the electron is

☐

@ left

☐

@ up

☐

@ down

☐

@ right



Question 3 25 pts

An electric field has a magnitude of 50,000 N/C

The force acting on an electron is:

(hint: 50,000 N/C means 50,000 N per 1 coulomb. $e = 1.6 \cdot 10^{-19} \text{C}$)

☐8 10^{-15} N☐3.2 10^{-23} N☐

50,000N

☐1.6 10^{-19} N

⋮

Question 4 25 pts

E is to electrostatics what _____-- is to gravity

☐

g

☐

q

☐

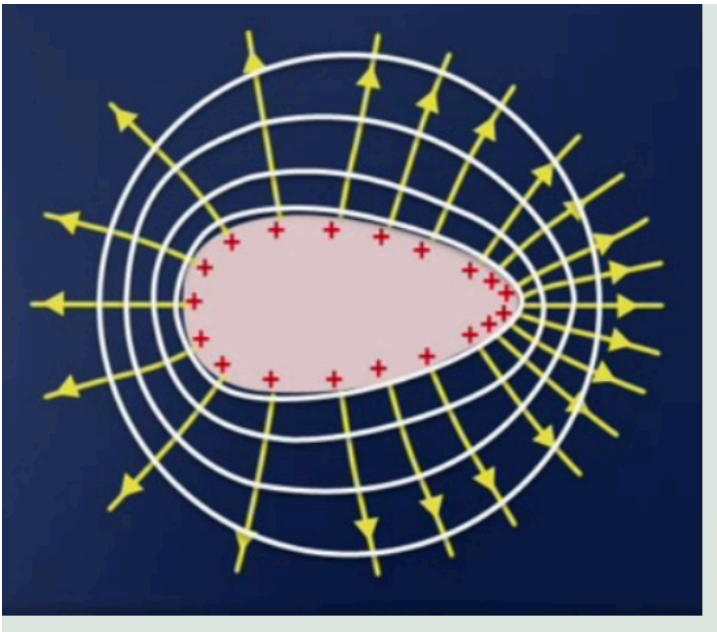
PE

☐

mass

⋮

Question 5 4 pts



Consider a metallic conductor (like a van graaf but with a weird shape) . IT is charged.

What is true

☐

the electric field is represented here like contour line around the structure



The electric field is stronger at the sharp edge



the electric field has a constant value inside the conductor



the electric is the same everywhere on the surface of the conductor



Question 6 3 pts

(MCAT)

An electron is accelerated over a distance d by an electric potential V . The electric potential applied to this electron is then increased by a factor of 4 and the electron is accelerated over the same distance d . The speed of the electron at the end of the second trial will be larger than at the end of the first trial by a factor of:

hint: work done on electron = change in KE

work = $e V = 0.5 m \text{ speed}^2$

if V is $\times 4$ what happens to speed^2 so what happens to speed ?

if you don't see it. Take $V=1V$ then $V=4V$. what happens to speed?



4



16



2



8



Question 7 2 pts

A certain 9 V battery is used as a power source to move a 2 C charge. How much work is done by the battery?



18J



4.5J

☐

36J

☐

9J

Not saved

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